Mongolia International Mathematics Competition 2026

Ulaanbaatar City, 22nd to 27th July 2026

Elementary Mathematics International Contest

Individual Contest

Time limit: 90 minutes

Information:

- You are allowed 90 minutes for this paper, consisting of 15 questions to which only numerical answers are required.
- Each question is worth 10 points. No partial credits are given. There are no penalties for incorrect answers, but you must not give more than the number of answers being asked for. For questions asking for several answers, full credit will only be given if all correct answers are found.
- Diagrams shown may not be drawn to scale.

Instructions:

- Write down your serial number, name, contestant ID and team's name on the answer sheet.
- Enter your answers in the space provided on the answer sheet.
- You are allowed to use HB, B, and 2B pencils and ball-point pens with black or blue ink.
- You cannot use instruments such as protractors, calculators and electronic devices, smart watches.
- At the end of the contest, you must hand in the envelope containing the question sheets, the answer sheet and all scrap paper.

English Version

1. During the Naadam festival, three warriors namely Andy, Ben and Charlie, competed in a 1 km horse race across the vast Mongolian grasslands. Each of their horses ran at a constant speed throughout the race. When Andy crossed the finish line, Ben was 40 meters behind him. When Ben crossed the finish line, Charlie was 25 meters behind him. How many meters behind Andy was Charlie at the exact moment Andy crossed the finish line?



2. At a party, there are 200 people, each of whom is either a liar (a person who always tells a lie) or a truth teller (a person who always tells the truth). During the party, some of them shake hands with each other. However, the room is too dark, so it is impossible to determine which pairs of people shook hands.
After the party, each person was asked:

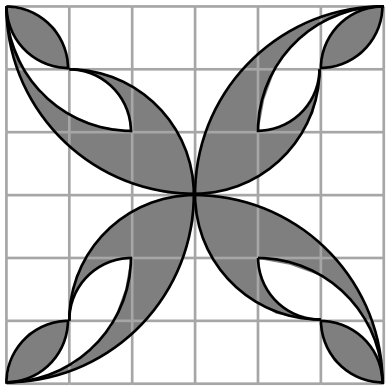
"How many truth tellers did you shake hands with?"

The answer the nth person gives is $n-1$, where $n=1, 2, 3, ..., 200$.
How many liars were at the party?

3. Given that $A=\dfrac{4051\times 4051}{2026\times 2025}-\dfrac{2025\times 2025}{2026\times 4051}-\dfrac{2026\times 2026}{2025\times 4051}$. What is the numerical value of A?

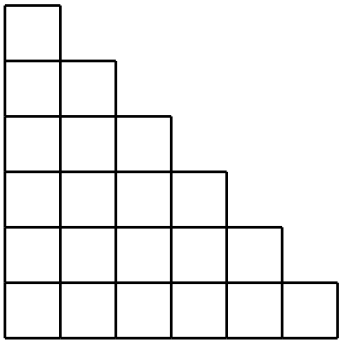
4. There are 20 students at the MIMC Cultural Night, where all the boys shake hands exactly once with each student. If all the girls don't shake hands with each other and there are exactly 145 handshakes in total, how many girls are attending this Cultural Night?

5. A 6 by 6 grid is shown in the diagram below, where each unit square has a side length of 7 cm. Each curve is a quarter of a circle, where the radius of this circle is equal to one or more side lengths of a unit square. What is the area, in cm^2, of the shaded region? (Note: Take $\pi=\dfrac{22}{7}$)

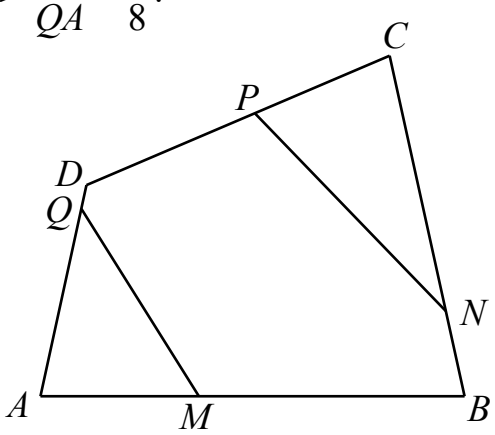


6. The 2026 FIFA World Cup consists of 48 teams divided into 12 groups of 4 teams each. In every group, each team plays every other team exactly once. A win earns 3 points, a draw earns 1 point, and a loss earns 0 points. After all group matches have been played, exactly k teams finish with 6 points. What is the largest possible value of k?

7. How many different rectangles (including squares) in different locations in the diagram below can be formed using the grid lines?
Note: For example, the unit squares count as 21 different rectangles.



8. As shown in the diagram below, $ABCD$ is a convex quadrilateral where points M, N, P and Q lie on sides AB, BC, CD and AD, respectively such that $\frac{AM}{MB}=\frac{3}{5}$, $\frac{BN}{NC}=\frac{1}{3}$, $\frac{CP}{PD}=\frac{4}{5}$ and $\frac{DQ}{QA}=\frac{1}{8}$.



If the area of $ABCD$ is 45 cm^2, what is the area, in cm^2, of hexagon $MBNPDQ$?

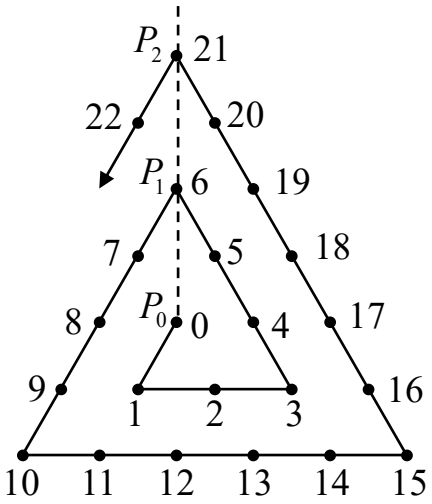
9. At the beginning, there are 5051 jars, each containing exactly one candy. In each turn, one candy is removed from each non-empty jar, then a new empty jar is added, and all the candies collected during this turn are placed into the newly added jar. After 5051 turns, what is the largest number of candies in any jar?

10. During a 40-day period, Andy saved money every day except on 4 skipped days. On day n, where $1 \le n \le 40$, if he did not skip that day, he saved n dollars. Otherwise, he saved nothing. The four skipped days formed two disjoint pairs of consecutive days, with at least one day between the two pairs. Andy never skipped more than two consecutive days. If Andy's total savings over the 40 days were 766 dollars, how many possible patterns of skipped days are there?

11. As shown in the diagram below, a frog jumps along a triangular spiral path, starting at point P_0 (which represents the number 0). The path consists of straight segments whose lengths increase by 1 unit after each turn: the first segment has length 1, the second has length 2, the third has length 3, and so on. Each jump moves the frog by exactly 1 unit along the path to the next numbered point. (e.g. jump 1 lands on 1, jump 2 lands on 2, jump 3 lands on 3, and so on.)

Following this pattern, the frog reaches point P_1 (position 6) after 6 jumps and point P_2 (position 21) after 21 jumps.

Let P_n denote the *n*-th point along the vertical line shown. How many jumps must the frog make to travel starting from P_0 to point P_{14}?



12. Consider all five-digit palindromic numbers, $\overline{abcba}$, where $a \neq 0$, which are divisible by 99. Let *M* and *m* be the largest and the smallest of these numbers. What is the value of $M + m$?

13. Three four-digit positive integers are written on a board. If every digit 2 written in the three numbers is replaced with 3, then the sum of the resulting numbers is 15301. If every digit 4 in the three original numbers is replaced with 7, then the sum of the resulting numbers is 16176. What is the sum of the three original numbers?

14. What is the fifth smallest positive integer *N* such that $2026 \times N$ has exactly 100 positive integer divisors?

15. Let x_1, x_2, …, x_{10} be a permutation of the numbers 1, 2, 3, 4, 5, 6, 7, 8, 9, 10. How many different permutations satisfy the following conditions?

$$x_1 < x_2 < x_3 < x_4 < x_5,$$
$$x_6 < x_7 < x_8 < x_9 < x_{10},$$
$$x_1 + x_2 + x_3 + x_4 + x_5 < x_6 + x_7 + x_8 + x_9 + x_{10}.$$